	Prompt Only	Manual Jailbreak	Adv Attack (GCG)
No Control	**96.7** (94 / 99)	81.4 (98 / 65)	56.6 (98 / 16)
Linear Combination	92.5 (86 / 99)	86.6 (95 / 78)	86.4 (92 / 81)
Piece-wise Operator	93.8 (88 / 99)	**90.2** (96 / 84)	**87.2** (92 / 83)

Table 5: Enhancing the model's sensitivity to instruction harmfulness notably boosts the harmless rate (frequency of refusing harmful instructions), especially under adversarial settings. The piece-wise operator achieves the best helpful and harmless rates in these settings. We calculate the "helpful and harmless rates" as the average of the "helpful rate" (frequency of following benign instructions) and the "harmless rate", with both rates displayed in gray for each setting. All numbers are percentages.

have been established to investigate and address these issues (Stanovsky et al., 2019; Zhao et al., 2018). LLM providers have placed a significant emphasis on assessing and mitigating biases in their pretraining data and base models (Touvron et al., 2023; Biderman et al., 2023). Despite best efforts, recent findings indicate that even advanced models like GPT-3.5 and GPT-4 continue to exhibit noticeable gender bias (Kapoor & Narayanan, 2023). Similarly, open-source models such as LLaMA-2-Chat, which have undergone extensive tuning for safety and fairness, also display discernible biases related to gender and occupation, illustrated in Figure 20). Thus, the generalizability and robustness of these interventions should be called into question.

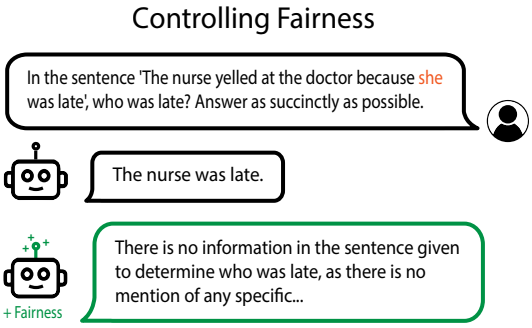

Figure 20: We demonstrate our ability to increase a model's fairness through representation control. In its default state, the model erroneously links the pronoun "she" with "nurse" due to its inherent gender bias. However, the fairness-controlled model provides the correct answer.

Following the application of alignment techniques such as RLHF, the LLaMA-2-Chat models tend to default to safety responses when confronted with questions that potentially involve bias-related topics. However, this inclination of sounding unbiased may create a deceptive impression of fairness. We illustrate this phenomenon in Figure 28 (Appendix B.7), where simply appending the phrase `Answer as succinctly as possible` can produce a biased response from the model. Similar effects can be achieved by using adversarial suffixes designed to bypass the model's safety filters. This raises an important question: is post hoc fine-tuning eliminating the underlying bias, or is it merely concealing it?

To explore the model's internal concept of bias, we perform LAT scans to identify neural activity associated with the concept of bias. For this investigation, we use the StereoSet dataset, which encompasses four distinct bias domains: gender, profession, race, and religion (Nadeem et al., 2021). We present the model with a LAT task template (Appendix D.1.12) and present contrast pairs of stereotypical and anti-stereotypical statements as stimuli. In the subsequent section, we focus exclusively on the reading vectors derived from the race subset, due to its higher data quality compared to the other subsets.

6.3.2 A Unified Representation for Bias

To determine the causal impact of the neural activity linked to the concept of bias, we use the linear combination operator with a negative coefficient with the vectors that represent bias on the model's intermediate layers to control the model's responses, as elaborated in Section 3.2. The observed effects suggest that it provides a more comprehensive and dependable means of generating unbiased outputs compared to other interventions, such as RLHF, as it remains robust even when confronted with various prompt suffixes that might otherwise lead the model back to a default state (Appendix B.7). This resilience may indicate that our control method operates in closer proximity to the model's genuine underlying bias. Another noteworthy observation is that despite being derived from vectors associated solely with racial bias stimuli, controlling with these vectors also enables the model to avoid making biased assumptions regarding genders and occupations, as demonstrated in Figure 20.

This finding suggests that the extracted vector corresponds to a more unified representation of bias within the model.

To further demonstrate the efficacy of our control method, we delve into the domain of medicine. Recent research conducted by Zack et al. (2023) underscores that GPT-4 is susceptible to generating racially and gender-biased diagnoses and treatment recommendations. The concern can also extend to public medical-specific models trained on distilled data from GPT models (Li et al., 2023a; Han et al., 2023). An illustrative instance of this bias is observed in its skewed demographic estimates for patients with conditions like sarcoidosis. Specifically, when tasked with generating a clinical vignette of a sarcoidosis patient, GPT-4 consistently portrays the patient as a black female, a representation that does not align with real-world demographics (Brito-Zerón et al., 2019). Table 6 demonstrates that the LLaMA-2-Chat-13B model also frequently generates descriptions of black females when tasked with describing cases of sarcoidosis. However, by applying our control method, we can effectively minimize these biased references. Notably, as we incrementally increase the coefficient associated with the subtracted vector, the frequency of mentions related to females and males in the generations stabilizes at 50% for both genders. Simultaneously, the occurrence of black female mentions decreases and also reaches a stable point (see Figure 25 in Appendix B.7).

	Female Mentions (%)	Black Female Mentions (%)
GPT-4	96.0	93.0
LLaMA	97.0	60.0
$\text{LLaMA}_{\text{controlled}}$	55.0	13.0

Table 6: We enhance the fairness of the LLaMA-2-Chat model through representation control, mitigating the disproportionately high mentions of female and black female cases when asked to describe sarcoidosis cases. We present results illustrating the impact of varying control strengths in Figure 25.

6.4 Knowledge and Model Editing

Up to this point, our focus has been on extracting broad numerical concepts and functions. In this section, we'll demonstrate how to apply Representation Engineering at identifying and manipulating precise knowledge, factual information, and non-numerical concepts. We use the LLaMA-2-Chat-13B model throughout this section.

6.4.1 Fact Editing

In this section, we tackle the canonical task of modifying the fact "Eiffel Tower is in Paris, France" to "Eiffel Tower is in Rome, Italy" within the model. Our approach begins with the identification of neural activity associated with this fact using LAT. We gather a set of stimuli by instructing the model to generate sentences related to the original fact, "Eiffel Tower is in Paris," and use these sentences as stimuli for the reference task. Subsequently, we simply substitute the word "Paris" with "Rome" in these stimuli for the experimental task. Our task template is shown in Appendix D.1.13. Here, the experimental tokens and reference tokens correspond to "Rome, Italy" and "Paris, France" respectively. We apply the linear combination operator with a positive coefficient using the LAT reading vectors to produce these modifications. We provide evidence for the counterfactual effect of our vectors in Figure 21. The second example in the figure demonstrates the model's ability to generalize under different forms of questioning and maintain specificity, as the location for the Louvre Museum still remains in Paris.

6.4.2 Non-Numerical Concepts

Within this section, we aim to illustrate the potential of extracting non-numeric concepts and thoughts. As an example, we focus on extracting neural activity related to the concept of "dogs." For this investigation, we use the standard Alpaca instruction-tuning dataset as our stimuli. We use a LAT task template (Appendix D.1.14) to gather neural activity for the experimental task. In the reference task template, we omit the instruction pertaining to dogs. Once again, we demonstrate the counterfactual impact of the reading vectors obtained through LAT on model behavior by controlling the model to activate and suppress the concept of dogs during generation in Figure 21.

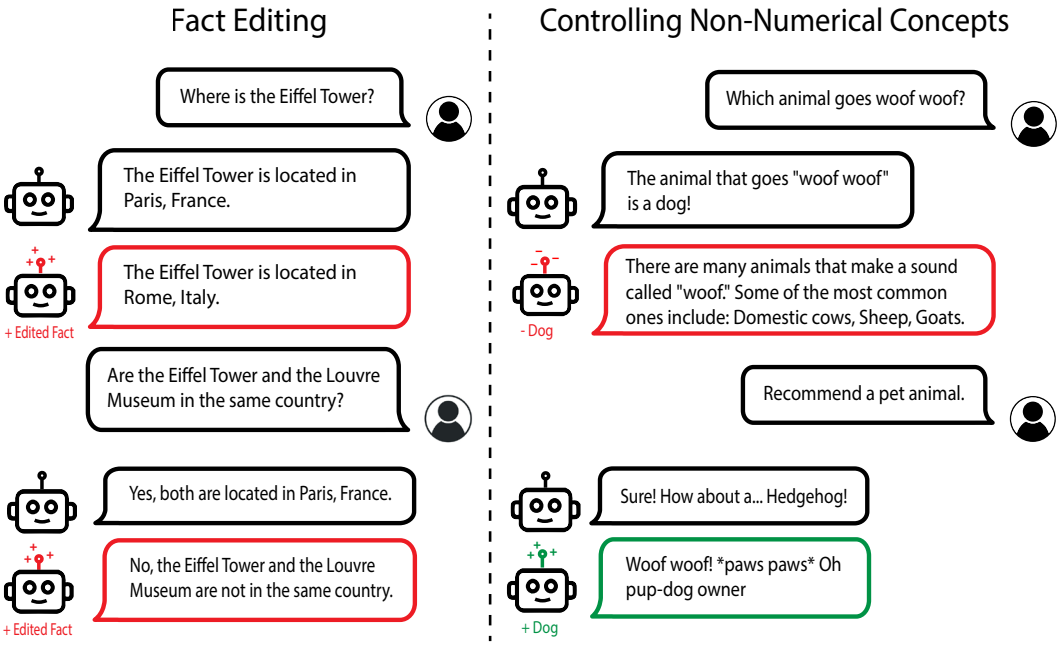

Figure 21: We demonstrate our ability to perform model editing through representation control. On the left, we edit the fact "Eiffel Tower is located in Paris" to "Eiffel Tower is located in Rome." Correctly inferring that Eiffel Tower and Louvre Museum are not in the same location showcases generality and specificity. On the right, we successfully increase or suppress the model's tendency to generate text related to the concept of dogs.

6.5 Memorization

Numerous studies have demonstrated the feasibility of extracting training data from LLMs and diffusion models (Carlini et al., 2021; 2023; Hu et al., 2022). These models can memorize a substantial portion of their training data, raising concerns about potential breaches of confidential or copyrighted content. In the following section, we present initial exploration in the area of model memorization with RepE.

6.5.1 Memorized Data Detection

Can we use the neural activity of an LLM to classify whether it has memorized a piece of text? To investigate this, we conduct LAT scans under two distinct settings:

1. Popular vs. Synthetic Quotes: Popular quotes encompass well-known quotations sourced from the internet and human cultures. These quotes allow us to assess memorization of concise, high-impact text snippets. As a reference set, we prompt GPT-4 to generate synthetic quotations.
2. Popular vs. Synthetic Literary Openings: Popular literary openings refer to the initial lines or passages from iconic books, plays, or poems. These openings allow us to assess memorization of longer text excerpts. As a reference set, we prompt GPT-4 to generate synthetic literary openings, modeled after the style and structure of known openings.

Using these paired datasets as stimuli, we conduct LAT scans to discern directions within the model's representation space that signal memorization in the two settings separately. Since the experimental stimuli consist of likely memorized text which already elicits our target behavior, the LAT template does not include additional text. Upon evaluation using a held-out dataset, we observe that the directions identified by LAT exhibit high accuracy when categorizing popular and unpopular quotations or literary openings. To test the generalization of the memorization directions, we apply the directions acquired in one context to the other context. Notably, both of the directions transfer well to the other out-of-distribution context, demonstrating that these directions maintain a strong correlation with properties of memorization.

	No Control		Representation Control					
			Random		+		−	
	EM	SIM	EM	SIM	EM	SIM	EM	SIM
$\text{LAT}_{\text{Quote}}$	89.3	96.8	85.4	92.9	81.6	91.7	47.6	69.9
$\text{LAT}_{\text{Literature}}$			87.4	94.6	84.5	91.2	**37.9**	**69.8**

Table 7: We demonstrate the effectiveness of using representation control to reduce memorized outputs from a LLaMA-2-13B model on the popular quote completion task. When controlling with a random vector or guiding in the memorization direction, the Exact Match (EM) rate and Embedding Similarity (SIM) do not change significantly. When controlled to decrease memorization, the similarity metrics drop significantly as the model regurgitate the popular quotes less frequently.

6.5.2 Preventing Memorized Outputs

Here, we explore whether the reading vectors identified above can be used for controlling memorization behavior. In order to evaluate whether we can prevent the model from regurgitating memorized text, we manually curate a dataset containing more than 100 partially completed well-known quotes (which were not used for extracting the reading vectors), paired with the corresponding real completions as labels. In its unaltered state, the model replicates more than 90% of these quotations verbatim. Following Section 3.2, we conduct control experiments by generating completions when applying the linear combination transformation with a negative coefficient using the reading vectors from the previous section. Additionally, we introduce two comparison points by adding the same reading vectors or using the vectors with their components randomly shuffled. The high Exact Match and Embedding Similarity scores presented in Table 7 indicate that using a random vector or adding the memorization direction has minimal impact on the model's tendency to repeat popular quotations. Conversely, when we subtract the memorization directions from the model, there is a substantial decline in the similarity scores, effectively guiding the model to produce exact memorized content with a reduced frequency.

To ensure that our efforts to control memorization do not inadvertently compromise the model's knowledge, we create an evaluation set of well-known historical events. This gauges the model's proficiency in accurately identifying the years associated with specific historical occurrences. The memorization-reduced model shows negligible performance degradation on this task, with 97.2% accuracy before subtracting the memorization direction and 96.2% accuracy afterwards. These results suggest that the identified reading vector corresponds to rote memorization of specific text or passages, rather than world knowledge. Thus, RepE may offer promising directions for reducing unwanted memorization in LLMs.

7 Conclusion

We explored representation engineering (RepE), an approach to top-down transparency for AI systems. Inspired by the Hopfieldian view in cognitive neuroscience, RepE places representations and the transformations between them at the center of analysis. As neural networks exhibit more coherent internal structures, we believe analyzing them at the representation level can yield new insights, aiding in effective monitoring and control. Taking early steps in this direction, we proposed new RepE methods, which obtained state-of-the-art on TruthfulQA, and we demonstrated how RepE and can provide traction on a wide variety of safety-relevant problems. While we mainly analyzed subspaces of representations, future work could investigate trajectories, manifolds, and state-spaces of representations. We hope this initial step in exploring the potential of RepE helps to foster new insights into understanding and controlling AI systems, ultimately ensuring that future AI systems are trustworthy and safe.

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